

Usual water-related behaviour and 'near-drowning' incidents in young adults

Abstract

Objective: To describe usual water-related behaviour and 'near-drowning' incidents in a cohort of young New Zealand adults.

Method: This was a cross-sectional study based on data collected as part of the Dunedin Multidisciplinary Health and Development Study, which is the study of a cohort (n=1,037) born between 1 April 1972 and 31 March 1973 in Dunedin, New Zealand. The data analysed were collected at age 21 (1993/94). Each study member was given a face-to-face interview using a structured questionnaire.

Results: Males reported a higher level of water confidence, exposure to risk behaviours, and exposure to unsafe locations, and more 'near-drowning' incidents, than the females, but protective behaviour did not differ. Males and females who were 'confident' in the water were more likely to be exposed to unsafe water locations, and water-confident males were more likely to drink alcohol before water activities, but not boating. A total of 169 'near-drowning' incidents were reported by 141 study members (63% males). 'Near-drowning' incidents were associated with unsafe swimming environments for males ($p<0.001$) and boating within two hours of consuming alcohol for females ($p=0.002$).

Conclusions: This study described usual water-related behaviour and has provided preliminary evidence of the factors associated with 'near-drowning' incidents among a high-risk age group. Larger case-control studies are required to further investigate risk factors for 'near-drowning'.

Implications for practitioners: Further investigation is required to determine the effectiveness of providing water skills acquisition in both safe and unsafe environments on 'near-drowning' experience.

(Aust N Z J Public Health 2005; 29: 238-43)

Pauline Gulliver and Dorothy Begg

Injury Prevention Research Unit, Dunedin School of Medicine, University of Otago, New Zealand

From a nation-wide sport and physical activity survey conducted in 1997/98, it was estimated that 36% of adult New Zealanders (those aged over 18 years) participated in swimming in the past 12 months. In addition, 25% were involved in fishing, 7% in surfing/body-boarding, 5% in yachting/sailing, 5% in aquarobics and 2% in rowing.¹

Considering the popularity of water-related activities in New Zealand, it is not surprising that water-related injuries rate highly in injury statistics. When combined, water-related activities ranked second only to rugby for sports-related claims accepted by a compulsory national insurance scheme.² Submersion injuries accounted for 5% of injury mortality in New Zealand between 1994 and 1998, which was the fourth-highest cause of injury mortality after motor vehicle, suicide, and falls.³ In 1996, submersion injuries accounted for 0.3% of public hospital discharges.⁴

There have been a few studies on the nature and circumstances of drowning incidents.⁵⁻⁷ For example, Langley and colleagues (2001) provided a detailed description of the causes of drowning obtained from three databases in New Zealand. Over the period 1980 to 1994, males and those aged 0-4 or 15-24 years were identified as having the highest rates of drowning. Boating was the leading cause of drowning, followed by swimming and other water sports.⁶ The authors concluded that, although there had been a significant reduction in drowning deaths, New Zealand compared poorly internationally. "In

particular, our unintentional drowning rate is twice that of Australia. For New Zealand to continue to make substantial progress in addressing its overall drowning rate, we need to ... strengthen our efforts in priority areas, in particular, those ... among preschoolers, adolescents and young adults" (p. 451).

Although some circumstances may differ between drowning and 'near-drowning' incidents, an examination of factors associated with 'near-drowning' incidents may further the understanding of risk factors for drowning and aid the development of interventions to target the priority areas identified by Langley et al (2001). One of the main strengths of 'near-drowning' studies is that those involved are alive and (potentially) able to describe what happened. There is, however, a lack of epidemiological evidence on the circumstances of, and activities associated with, 'near-drowning' incidents. Of the studies that have been undertaken, one was in the United States⁸ and another in Australia.⁹ Although both studies collected information on drowning and 'near-drowning' incidents, neither differentiated their findings by the activities associated with either outcome. Thus, it was not determined if these two outcomes shared common precedents or causal factors for injury. Other authors have focused on the location of 'near-drowning' incidents (e.g. see Jensen et al.¹⁰), but the usefulness of such studies is limited by a lack of descriptive information on the activity undertaken at the time of the injury event. There are no published reports describing

Submitted: March 2004

Revision requested: August 2004

Accepted: February 2005

Correspondence to:

Dr Dorothy Begg, Injury Prevention Research Unit, Dunedin School of Medicine, University of Otago, PO Box 913, Dunedin, New Zealand.
Fax: +64 3 479 8337; e-mail: dorothea.begg@ipru.otago.ac.nz

the circumstances associated with 'near-drowning' incidents in New Zealand.

The aim of the present study was to describe the usual water-related behaviour and 'near-drowning' incidents in a cohort of young New Zealand adults. In particular, we sought to describe water-related factors such as water confidence, exposure to risk environments, safety and risk behaviours, and the association between these factors and a 'near-drowning' experience.

Methods

The data presented were collected as part of the Dunedin Multidisciplinary Health and Development Study, which is a longitudinal study of a cohort (n=1,037) born between 1 April 1972 and 31 March 1973 at the only obstetric hospital in Dunedin, New Zealand. A full description of the cohort and the DMHDS is provided elsewhere.¹¹ Briefly, since the age of three years the cohort has been assessed every two years up to 15 years of age, and then again at ages 18, 21 and 26 years. The data analysed in this paper were collected at age 21 (1993/94), when 97% of the sample were followed up.

Each study member was given a face-to-face interview by a trained interviewer using a structured questionnaire. The questions concerning water safety were divided into five sections:

1. Water confidence.
2. Protective behaviours – life jackets, lifesaving training, cardio-pulmonary resuscitation training.
3. Risk behaviours – involvement in water-based activities within two hours of consuming alcohol.
4. Exposure to risk – location of participation in water-related activities.
5. Experience of 'near-drowning' incidents.

Water confidence

To assess water confidence, study members were asked if they could swim 50 metres freestyle non-stop in a warm, heated swimming pool; tread water 20 metres off shore in a deep lake with a wind blowing that produced a 10-centimetre chop. They also rated how they felt about performing these tasks (0=comfortable, 1=anxious, 2=very anxious). Those who were comfortable at performing both tasks were coded as 'confident'. Those who were anxious or very anxious at performing either task were coded as 'anxious'.

Protective behaviours

The protective behaviours assessed were cardio-pulmonary resuscitation (CPR), lifesaving training, and the use of flotation devices. These protective behaviours were included as they indicate an overall safe attitude towards the water. Each study member was asked if they had received lifesaving training (response options: yes/no) or were certified in CPR. They were also asked how often they wore a life jacket (or similar flotation device) when they went boating (response options were on a four-point scale: never – always).

Risk behaviours

The risk behaviours included how many times in the past year they had been boating within two hours of drinking alcohol, and how many times they had done other water-related activities (such as wind-surfing, water skiing, swimming, etc.) within two hours of drinking alcohol. Responses were dichotomised: '0' indicating zero times, '1' indicating at least once.

Exposure to risk

From the following list of locations, study members were asked to indicate where they had participated in water-based activities within the past three years:

- On a beach (between the flags).
- On a beach (patrolled, but not between the flags).
- On a beach (unpatrolled).
- In a river.
- In a lake.
- In a public pool.
- In a private pool.
- In a harbour.
- In the sea.

To classify the above localities in terms of risk, a reliability survey was conducted with staff members at the Injury Prevention Research Unit, University of Otago. Each staff member (n=11) was asked to rate each of the localities on a four-point scale ('very safe' – 'very unsafe'). 'Unsafe' localities (those that at least 80% of staff members rated as unsafe) were the harbour and the open sea. 'Safe' localities (those that at least 80% of staff members rated as safe) were a public swimming pool, at the beach between the flags, and a private swimming pool. The remaining localities were classified as neutral. Each study member's exposure to risk was classified as: those that only participated in water-based activities in 'safe' localities were classified as 'safe'; those that participated in water-based activities in safe and neutral localities were classified as 'neutral'; and those that participated in water-based activities in any of the 'unsafe' localities were classified as 'unsafe'.

'Near-drowning' incidents

Participants were asked whether, in the last three years, they had any experience on, in, or near the water when they were afraid they might drown. Although the majority of these incidents may not concur with a technical definition of 'near-drowning', they appeared as such to the study members. Hence, they have been recorded as 'near-drowning', but this term has been placed in quotation marks to highlight this discrepancy. A brief description was recorded for each incident to provide locality and activity data.

Analyses

Quantitative data were analysed using SAS. Frequencies are presented for each of the above variables, and the chi-squared statistic was used to examine differences in explanatory variables

Table 1: Water confidence, protective behaviours, risk behaviours and exposure to risk.

	Males n (%)	Females n (%)	χ^2	<i>p</i>
Water confidence				
Swim 50 metres in a warm, shallow pool				
Yes	421 (89)	364 (81)	14.22	<0.001
No	50 (10)	88 (19)		
How do you feel about this swimming task?				
Comfortable	355 (84)	291 (80)	2.58	0.27
Anxious	61 (14)	67 (18)		
Very anxious	5 (1)	6 (2)		
Tread water 20m off shore in a deep lake with a wind blowing that gives a 10cm chop?				
Yes	451 (97)	421 (93)	6.99	0.01
No	15 (3)	32 (7)		
How do you feel about this treading water task?				
Comfortable	307 (68)	199 (47)	50.41	<0.001
Anxious	122 (27)	152 (36)		
Very anxious	22 (5)	70 (17)		
Protective factors				
Have you been certified in CPR?				
Yes	290 (61)	298 (65)	1.41	0.24
No	183 (39)	160 (35)		
Have you had lifesaving training?				
Yes	234 (49)	214 (47)	0.76	0.38
No	240 (51)	246 (53)		
Risk behaviours				
In the past three years, have you been boating?				
Yes	345 (73)	262 (57)	26.52	<0.001
No	129 (27)	200 (43)		
Have you been boating within two hours of drinking alcohol? ^a				
Yes	72 (21)	25 (10)	14.26	<0.001
No	274 (79)	238 (90)		
Have you been involved in other water-related activities within two hours of drinking alcohol? ^a				
Yes	106 (23)	38 (9)	35.24	<0.001
No	353 (77)	404 (91)		
When you go boating, how often do you wear a life jacket?				
Never	57 (17)	33 (13)	12.56	0.006
Sometimes	61 (18)	31 (12)		
Nearly always	52 (15)	28 (11)		
Always	174 (51)	170 (65)		
Exposure to risk				
In the past three years I have been involved in water-based activities in these area(s):				
Beach (between the flags)	265 (55)	274 (59)	1.01	0.29
Patrolled beach (not between the flags)	187 (39)	137 (30)	9.92	0.002
Unpatrolled beach	254 (52)	189 (41)	15.08	<0.001
River	296 (61)	214 (46)	24.54	<0.001
Lake	297 (61)	233 (50)	14.24	<0.001
Public pool	300 (62)	337 (73)	10.03	0.002
Private pool	186 (38)	168 (36)	0.82	0.36
Harbour	139 (29)	80 (17)	18.82	<0.001
Open sea	172 (36)	116 (25)	13.73	<0.001
'Exposure to risk' classification				
Safe	50 (11)	107 (23)	34.72	<0.001
Neutral	193 (41)	198 (43)		
Unsafe	231 (49)	157 (34)		

Note:

(a) Each of these questions were prefixed with the statement: In the past year...

(protective factors, risk factors, and exposure to risk) by gender, water confidence, and 'near-drowning' incidents. Descriptive data for 'near-drowning' incidents were used to describe the locality where the incident occurred and the activity undertaken at the time.

Results

The majority (85%) of the study members reported they were able to swim 50 metres non-stop in a warm shallow pool. Eighty-two per cent felt comfortable about performing this task. While 95% of study members also reported being able to tread water 20 metres off shore in a deep lake with a wind that created a 10-centimetre chop, only 58% reported that they felt comfortable about performing this task. Sixty-three per cent of study members reported that they had been trained in CPR, and 48% indicated that they had received lifesaving training. Nearly two-thirds (64%) reported boating in the three years prior to the interviews. Of those who had gone boating, 16% reported that they had done so within two hours of drinking alcohol. Sixteen per cent of study members reported being involved in other water-based activities within two hours of drinking alcohol in the year prior to the interviews. Fifty-seven per cent of those who had gone boating reported that they always wore a life jacket when boating, 30% reported that they 'sometimes' or 'never' wore a life jacket.

All study members had participated in water-based activities in at least one of the localities listed in the three years prior to the

interviews. Over half had participated in activities at the beach (between the flags) (57%), in rivers (54%), or in lakes (56%).

Gender differences for each of the descriptive variables are presented in Table 1. Compared with the females, the males reported a higher level of water confidence, were more likely to engage in risk behaviours, and had higher exposure to unsafe water locations, but did not differ on protective factors. Because of these significant differences, the remaining analyses have been conducted by gender.

To investigate the relationship between water confidence and water-related activities, the confidence measure was analysed for each of the risk behaviours, exposure to risk, and 'near-drowning' experience (see Tables 2 and 3 for males and females, respectively). Males classified as 'confident' were more likely to be involved in water-based activities (other than boating) within two hours of consuming alcohol, and to be involved in water-based activities in unsafe localities (see Table 2). Females classified as 'confident' were more likely to be involved in water-based activities in unsafe localities (see Table 3). Those classified as 'confident' for either gender were not more likely to go boating within two hours of consuming alcohol, go boating without a life jacket, or report a 'near-drowning' incident.

'Near-drowning' incidents

A total of 169 'near-drowning' incidents were reported by 141 study members. Females accounted for 37% (n=52) and males

Table 2: The relationship between water confidence and risk behaviours in males.

Explanatory variables	Males – confidence			
	Comfortable	Anxious	χ^2_{2a}	<i>p</i>
Boating within two hours of consuming alcohol				
Yes	64 (21)	8 (22)		
No	244 (79)	28 (78)	0.04	0.84
Wore a life jacket while boating				
Never/sometimes	105 (34)	13 (37)		
Most of the time/always	202 (66)	22 (63)	0.12	0.73
Other ^b water activities within two hours of consuming alcohol				
Yes	102 (25)	4 (8)		
No	307 (75)	44 (92)	6.64	0.01
'Exposure to risk' classification				
Safe	36 (8)	14 (25)		
Neutral	172 (41)	20 (36)		
Unsafe	208 (50)	22 (39)	13.99	0.001
Experienced a 'near drowning' episode in the previous three years				
Yes	78 (19)	9 (16)		
No	338 (81)	47 (84)	0.24	0.63

Notes:

(a) *df*=1 except for 'Exposure to Risk' classification where *df*=2.

(b) 'Other' = not including boating.

Table 3: The relationship between water confidence and risk behaviours in females.

Explanatory variables	Females – confidence			
	Comfortable	Anxious	χ^2_{2a}	<i>p</i>
Boating within two hours of consuming alcohol				
Yes	20 (9)	4 (10)		
No	199 (91)	36 (90)	0.03	0.86
Wore a life jacket while boating				
Never/sometimes	51 (23)	11 (28)		
Most of the time/always	168 (77)	28 (72)	0.44	0.51
Other ^b water activities within two hours of consuming alcohol				
Yes	31 (9)	5 (5)		
No	309 (91)	87 (95)	1.29	0.26
'Exposure to risk' classification				
Safe	60 (17)	45 (44)		
Neutral	155 (44)	37 (36)		
Unsafe	134 (38)	21 (20)	32.91	<0.001
Experienced a 'near drowning' episode in the previous three years				
Yes	40 (11)	12 (12)		
No	309 (89)	91 (88)	0.003	0.96

Notes:

(a) *df*=1 except for 'Exposure to Risk' classification where *df*=2.

(b) 'Other' = not including boating.

63% (n=117). Sixty (36%) of these incidents occurred while swimming, 48 (28%) while surfing, and 39 (23%) occurred while boating. A small number of 'near-drowning' incidents occurred while walking next to or in water (n=8, 5%), diving (n=7, 4%), windsurfing (n=4, 2%) or water skiing (n=3, 2%).

The relationship between involvement in a 'near-drowning' incident (yes/no) and the following explanatory factors was examined: water confidence, protective factors (certified in CPR, lifesaving training, life-jacket use), risk behaviours (boating and alcohol, other-water activity and alcohol), and exposure to risk environments (safe, neutral, unsafe locations). For the males, the only explanatory factor significantly associated with being involved in a 'near-drowning' incident was exposure to risk environments (74% vs. 43%, $\chi^2 = 27.54$, $df=2$, $p<0.001$). For the females, the only significant factor was that those who reported a 'near-drowning' incident were more likely to report boating within two hours of consuming alcohol (24% vs. 7%, $\chi^2 = 9.53$ $df=1$, $p=0.002$).

Location and activity of the 'near-drowning' incidents

Sixty study members reported 'near-drowning' incidents that occurred while they were swimming. The largest proportion of these incidents occurred at a beach and when they were caught in a rip or knocked over by waves (reported in 16 (27%) of cases). Almost half (n=20, 42%) of the surfing incidents occurred when they had been 'dumped by waves' (submerged). These people reported being held under, or dragged out by the waves and 'dumped'. Incidents that occurred while surfing at the beach were the largest contributor to 'near-drowning' incidents in males (n=35).

Boating incidents involved rafts (n=11), boats (n=10), canoes/kayaks (n=12) and yachts (n=6). White-water rafting accounted for the majority of rafting incidents (n=7, 70%). Fifteen (38%) boating 'near-drowning' incidents occurred at sea.

Of the other 'near-drowning' incidents, eight occurred when walking next to or in the water, and four occurred while attempting a river crossing. Two of the seven diving incidents occurred as a result of the diver becoming fatigued, or because they had gone too deep in their dive.

Discussion

As far as the authors are aware, this is the first study to describe water-related activities among young adults who are the age group at the highest risk of drowning, and to investigate the relationship between usual water-related behaviours and a 'near-drowning' incident.

The main results from this study showed that, compared with the females, males reported a significantly higher level of confidence in the water, higher exposure to risk behaviours, were more likely to be exposed to unsafe water locations, and were more likely to report a 'near-drowning' incident, but did not differ on the measures of protective behaviour. The relationship between

water confidence and other water-related behaviours showed that both males and females who were 'confident' in the water were more likely to participate in activities in unsafe water locations, such as the harbour and the open sea, than those who were not so confident. Also, the males who were confident in the water were more likely to engage in water-related activities (but not boating) after drinking alcohol. Of the wide range of usual water-related behaviours investigated, only exposure to unsafe environments (males) and boating within two hours of consuming alcohol (females) were related to experiencing a 'near-drowning' incident.

These results concur with other studies on drowning or 'near-drowning' events, where the majority of 'near-drowning' events involved males.¹²⁻¹⁴ Howland and colleagues found that men were more likely to rate themselves as 'excellent' or 'very good' swimmers even when they had undertaken no formal swimming lessons. They suggested that men may over-estimate their swimming ability and thus place themselves in riskier aquatic situations than women.¹⁴ The present results also support Howland et al.'s hypothesis that the reason for the difference in drowning experience between males and females is that males have a greater exposure to aquatic environments, to high-risk environments, and that they drink alcohol more frequently than women when on or near the water.¹⁴

In the present study, among those who engaged in boating and other water-related activities, alcohol was more likely to be involved among the males than the females. However, alcohol associated with water activity was not shown to be associated with experience in a 'near-drowning' incident for the males, but it was for the females. One of the limitations of this study was that the amount of alcohol consumed was not requested, neither was alcohol consumption immediately prior to the 'near-drowning' incident. Determining the involvement of alcohol in drowning and 'near-drowning' events is fraught with difficulties.¹⁵ However, there is evidence of a link between alcohol consumption and untoward aquatic events. Smith et al. (2001)¹⁶ conducted a case-control study of drinking and recreational boating fatalities. They reported a strong positive association between blood alcohol concentration (BAC) and the relative risk of drowning. Furthermore, the relative risk increased markedly as BAC increased, from 1.3 at a BAC of 10mg/dL to 52 at a BAC of 250 mg/dL. The results of the present study, therefore, should be viewed with caution, until further investigation of these factors has been undertaken.

The investigation of the relationship between water confidence and exposure to risk showed that for both males and females, confidence in the water was significantly associated with increased exposure to unsafe locations, but not to a 'near-drowning' incident. For the males, however, exposure to unsafe environments was associated with an increased risk of a 'near-drowning' incident. This suggests that, for the males, those who were more confident in the water may be more likely to have a 'near-drowning' experience, not because of their confidence in the water, but because of the unsafe locations in which they take part in water-related activities.

The current study showed that swimming accounted for a high proportion of reported 'near-drowning' incidents, and two-thirds of swimming incidents occurred at a beach. Twenty-eight per cent of the incidents were surfing related, more than 40% of these were described as being 'dumped' and dragged under by waves. The most frequently reported factors associated with a 'near-drowning' incident, irrespective of the activity undertaken at the time, was being knocked over by a wave or being caught in a rip. This suggests that water skills acquisition and injury prevention strategies should be aimed at seawater environments, as well as swimming pools, to realistically reflect the locations in which participants of water-based recreational activities find themselves in difficulties.^{17,18}

This study has provided preliminary evidence on the usual water-related behaviours and the association of these factors with 'near-drowning' incidents. Larger case-control studies are required to determine if the variables identified are risk factors for 'near-drowning'. Beach-goers (swimmers and surfers) and boaters would provide a useful sampling frame for such a study in New Zealand.

Acknowledgements

This research was funded by the Accident Rehabilitation and Compensation Insurance Corporation (ACC) of New Zealand. The Dunedin Multidisciplinary Health and Development Research Unit is funded by the Health Research Council of New Zealand and the US National Institute of Health, Grant No M4-45070. The Injury Prevention Research Unit (IPRU) is funded jointly by the Health Research Council of New Zealand and the ACC. The authors wish to acknowledge the assistance of Mr Warren Dukes (Department of Physical Education, University of Otago) for his advice on the water-related measures for this study. The ongoing support of Dr Phil Silva and Associate Professor Richie Poulton, past and present directors of the DMHDS, respectively, is also acknowledged, as well as Professor John Langley, Dr Sheila Williams and Mr Shaun Stephenson for their assistance and comments on an earlier draft of this paper. Last, but certainly not least, we wish to thank the members of the DMHDS cohort whose ongoing co-operation has made this study possible.

References

1. Business and Economic Research Limited. *The Growing Business of Sport and Leisure, The Impact of the Physical Leisure Industry in New Zealand, An Update to the 1998 Report* [formerly the Push Play Facts, Hillary Commission]. Wellington (NZ): SPARC Sport and Recreation New Zealand; 2000.
2. Accident Compensation Corporation. *ACC Injury Statistics 2002*. 2nd ed. Wellington (NZ): ACC; 2003.
3. Dow NA, Stephenson SCR, Casey DM. *Injury Causes by Age: IPRU Factsheet Number 29*. Dunedin (NZ): Injury Prevention Research Unit, University of Otago; 2003.
4. Wright C, Langley J, Allnatt D. *Causes of Injury: Fact Sheet Number 15*. Dunedin (NZ): Injury Prevention Research Unit, University of Otago; 1999.
5. Fingerhut LA, Cox CS, Warner M. International comparative analysis of injury mortality, findings from the ICE on injury statistics. *Adv Data*. 1998;303:1-20.
6. Langley JD, Warner M, Smith GS, Wright C. Drowning-related deaths in New Zealand, 1980-1994. *Aust N Z J Public Health*. 2001;25(5):451-7.
7. Mackie IJ. Patterns of drowning in Australia, 1992-1997. *Med J Aust*. 1999;171(11-12):587-90.
8. Quan L, Gore EJ, Wentz K, Allen J, Novack AH. Ten-year study of pediatric drownings and 'near-drownings' in King Country, Washington. *Pediatrics*. 1989;83(6):1035-40.
9. Patrick M, Bint M, Pearn J. Salt water drowning and 'near-drowning' accidents involving children – a five year population study in south-east Queensland. *Med J Aust*. 1979;1:61-4.
10. Jensen LR, Williams SD, Thurman DJ, Keller PA. Submersion injuries in children younger than five years in urban Utah. *West J Med*. 1992;157(6):641-4.
11. Silva PA, Stanton WR. From child to adult. In: *The Dunedin Multidisciplinary Health and Development Study*. Auckland (NZ): Oxford University Press; 1996.
12. Quan L, Cummings P. Characteristics of drowning by different age groups. *Injury Prev*. 2003;9:163-8.
13. Lindholm P, Steensberg J. Epidemiology of unintentional drowning and near drowning in Denmark in 1995. *Injury Prev*. 2000;6:29-31.
14. Howland J, Hingson R, Mangione TW, Bell N, Bak S. Why are most drowning victims men? Sex differences in aquatic skills and behaviours. *Am J Public Health*. 1996;86:93-6.
15. Warner M, Smith GS, Langley JD. Drowning and alcohol in New Zealand: what do the coroner's files tell us? *Aust N Z J Public Health*. 2000;24(4):387-90.
16. Smith GS, Keyl PM, Hadley JA, Bartley CL, Foss RD, Tolbert WG, et al. Drinking and recreational boating fatalities. A population-based case-control study. *J Am Med Assoc*. 2001;286(23):2974-80.
17. Hartung GH, Goerberg DA. Ocean sports. In: Caine D, Caine C, Linder KJ, editors. *Epidemiology of Sports Injury*. Champaign (IL): Human Kinetics; 1996.
18. Loneon M, Donaldson A, Ansems H. We will fight them on the beaches: surfing safety – a review of literature, risk factors and cultural factors in the Australian context. *Proceedings of the 4th World Conference on Injury Prevention and Control*; 1998 May; Amsterdam, Netherlands.